

Sol (1):- Given a continuous-time LTI system's difference eqⁿ,

$$\frac{d^2 y(t)}{dt^2} - \frac{dy(t)}{dt} - 2y(t) = x(t) \quad \text{--- (1)}$$

(a) As we know $x(t) \xrightarrow[\text{inverse}]{\mathcal{L}} X(s)$ with ROC = R where $s = \sigma + j\omega$

$$X(s) \triangleq \int_{-\infty}^{\infty} x(t) \cdot e^{-st} dt \quad \text{--- (2)}$$

Property:- Differentiation in the time domain

① mark: $\frac{d}{dt} x(t) \xrightarrow{\mathcal{L}} sX(s)$, with ROC containing R

$\frac{d^n}{dt^n} x(t) \xrightarrow{\mathcal{L}} s^n X(s)$, with ROC containing R

Taking Laplace on both sides in equation (1), we get

$$s^2 Y(s) - sY(s) - 2Y(s) = X(s) \quad \text{--- (3)}$$

$$Y(s) [s^2 - s - 2] = X(s)$$

$$H(s) = \frac{Y(s)}{X(s)} = \frac{1}{s^2 - s - 2} \quad \text{--- (5) --- 1 mark}$$

$$H(s) = \frac{1}{(s+1)(s-2)} = \frac{A}{s+1} + \frac{B}{s-2} \quad \text{--- (6) --- Partial fraction}$$

$$\Rightarrow 1 = A(s-2) + B(s+1) \quad \text{--- (7)}$$

Put $s = 2$, we get

$$\Rightarrow 1 = A(2-2) + B(2+1)$$

$$= 0 + 3B = 3B$$

$$\Rightarrow B = 1/3$$

Now, Put $s = -1$

$$\Rightarrow 1 = A(-1-2) + B(-1+1)$$

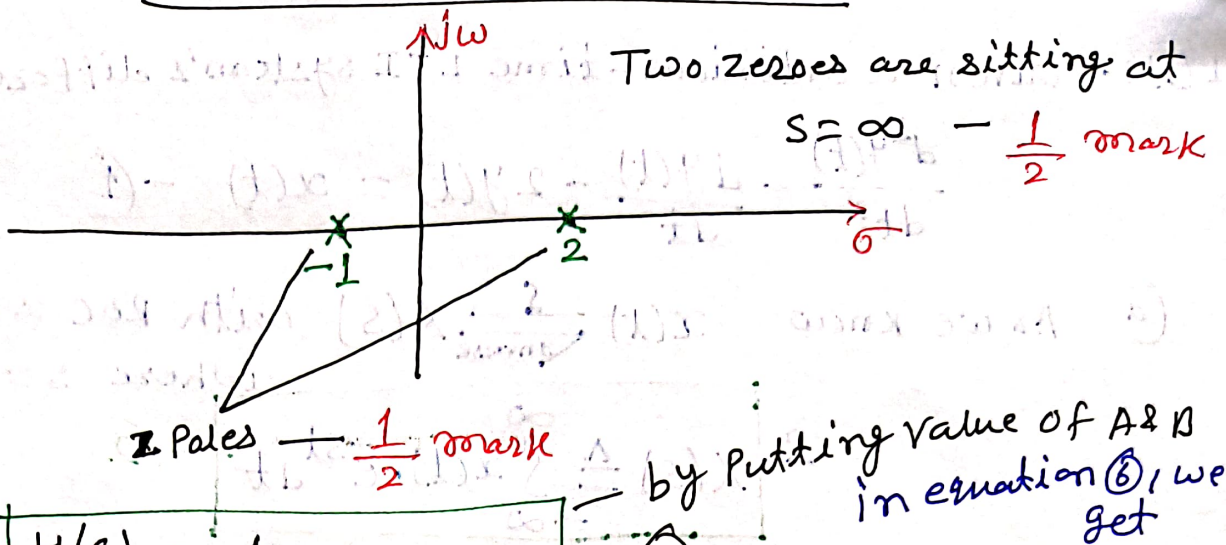
$$\Rightarrow 1 = A(-3) + B \times 0 = -3A$$

$$\therefore A = -1/3$$

$$\left. \begin{array}{l} A = -1/3 \\ B = 1/3 \end{array} \right\}$$

Sketching Pole-zero Pattern

(2)



$$H(s) = \frac{1/3}{s-2} - \frac{1/3}{s+1}$$

by putting value of A & B in equation (6), we get

1 mark

(b) (i) stability

→ An LTI system is stable if & only if the ROC of its system function $H(s)$ includes the entire $j\omega$ axis (i.e. $\text{Re}(s)=0$)

This will only be possible if the ROC for $H(s)$ has to be

$$-1 < \sigma < 2$$

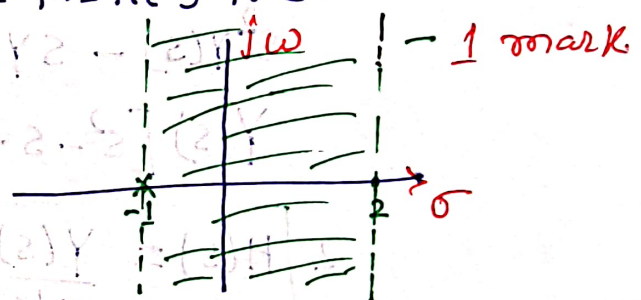
Therefore

$$H(s) = \frac{1/3}{s-2} - \frac{1/3}{s+1}$$

Taking Inverse Laplace Transform

$$h(t) = -\frac{1}{3}e^{2t}u(-t) - \frac{1}{3}e^{-t}u(t)$$

→ 1 mark



$$\begin{cases} e^{s t} \cdot u(t) = \frac{1}{s + \alpha}, \sigma > -\alpha \\ -e^{s t} \cdot u(-t) = \frac{1}{s + \alpha}, \sigma < -\alpha \end{cases}$$

(ii) Causality

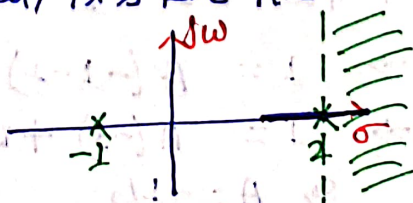
→ For a system with a rational system function, causality of the system is equivalent to the ROC being the right-half plane to the right of the rightmost pole.

→ For system should be causal, its ROC has to be $\text{Re}(s) > 2$

So, ROC: $\sigma > 2$

Right most Pole is $s = 2$

→ 1 mark



So ROC $\sigma > 2$ $H(s) = \frac{1/3}{s-2} - \frac{1/3}{s+1}$

Therefore

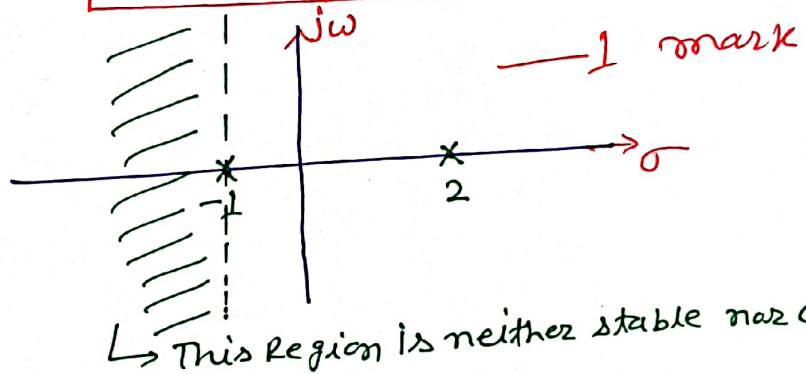
$h(t)$ after taking inverse Laplace transform is

$$h(t) = \frac{1}{3} e^{2t} u(t) - \frac{1}{3} e^{-t} u(t) \quad \text{--- 1 mark}$$

Qiii) Neither stable nor Causal

→ An LTI system is neither stable nor Causal if the ROC does not contain the $j\omega$ axis & it also does not have ROC, right to the right most pole.

So ROC has to be $\sigma < -1$ & $\text{Re}(s) < -1$



So, $H(s) = \frac{1/3}{s-2} - \frac{1/3}{s+1}$

Taking inverse Laplace transform

$$h(t) = -\frac{1}{3} \cdot e^{2t} u(-t) - \left(-\frac{1}{3} e^{-t} u(-t) \right)$$

$$h(t) = -\frac{1}{3} \cdot e^{2t} u(-t) + \frac{1}{3} \cdot e^{-t} u(-t) \quad \text{--- 1 mark}$$

~~proof of~~ $x(t) = e^{at} u(t)$ $\xrightarrow{\text{Laplace}} \frac{1}{s-a}$

$$X(s) = \int_{-\infty}^{\infty} e^{at} u(t) \cdot e^{-st} dt$$

$$= \int_0^{\infty} e^{at} \cdot e^{-st} dt$$

$$= \int_0^{\infty} e^{(s+a)t} dt = \left[\frac{e^{(s+a)t}}{(s+a)} \right]_0^{\infty}$$

$$= \frac{[e^{-\infty} - e^0]}{-(s+a)} = \frac{-1}{-(s+a)} = \frac{1}{s+a} \quad \boxed{\sigma > -a}$$

$-\frac{1}{2}$ if not done